

Show that the equation $x^3 + 2y^3 = 4z^3$ does not have any solutions in positive integers.

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$$x^3 + 2y^3 \equiv 4z^3 \equiv 0 \pmod{2}$$

$$2 \mid 2$$

$$x = 2x' \quad (x' \geq 1)$$

$$8x'^3 + 2y^3 = 4z^3$$

$$4x'^3 + y^3 \equiv 2z^3 \pmod{2}$$

$$2 \mid y$$

$$y = 2y' \quad (y' \geq 1)$$

$$4x'^3 + 8y'^3 = 2z^3$$

$$2x'^3 + 4y'^3 \equiv z^3 \equiv 0 \pmod{2}$$

$$2 \mid z$$

$$z = 2z' \quad (z' \geq 1)$$

$$\left. \begin{aligned} 2x'^3 + 4y'^3 &= 8z'^3 \\ 4x'^3 + 2y'^3 &= 4z'^3 \end{aligned} \right\}$$

If (x, y, z) is a solution to the given equation, then $(\frac{x}{2}, \frac{y}{2}, \frac{z}{2})$ is also a solution.

By well-ordering principle, we can let a solution (a, b, c) such that a is minimal.

Therefore if (a, b, c) is a solution, then $(\frac{a}{2}, \frac{b}{2}, \frac{c}{2})$ is also a solution. Since $\frac{a}{2} < a$ and this contradicts with the condition that a is least, the assumption contradicts.